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Faculty of Computing

CS-212: Object-Oriented Programming

Class: BESE-16B

Instructor: Dr. M. Daud Abdullah Asif

Final Report on the Semester Project

“PennyPoint: A Money-Spending Tracking App”

By: Maryam Ateeq (549392) and M. Shahmeer Shaheedi (565007)



About the Project and its Objectives

It is often difficult to track your monthly expenses, no matter who you are or where you're from. Not keeping a proper record of money inflow/outflow makes it difficult to visualize wealth. This leads to poor spending habits because even traditional noting methods are ineffective.

To solve this problem, we developed PennyPoint, an app that focuses on tracking the expenses of its users, helping them budget and save money. The application will allow users to easily record their daily expenses and organize them into categories.

The app doesn't just intend to help users classify their finances, but to also motivate them to abstain from wasting money. It also includes features like budget prediction and a no-spend streak tracker to encourage better financial habits.

Scope, Features, and Functionality

PennyPoint, after the initial designing process and implementation via code, comes out to incorporate the following features:

- **Add Expense:**
Users will be able to enter the amount of money they spend on the daily by entering details such as the amount, category of the expenditure, and date. This helps them see not only how much they spent from their allowance, but also where they've spent it, and when.
- **Manage Expenses:**
Users of PennyPoint will be able to manipulate their expenses even after they've added them into the app. For example, they will be able to edit the amount allocated to a certain category, delete an expense and its category altogether, search the expense by category, and sort the expenditures by amount.
- **Budget Prediction:**
Users can set a monthly budget, and the system will track their average daily spending and predict if/when they will start to exceed their monthly budget. This feature will be updated every time a new expense is allocated/entered.
- **Planner:**
This feature will analyze the user's activity and will track a month of unique spending days, and then will be able to enter a plan based on the user's spending habits. This includes suggestions and advice.
- **View Summary:**
The user will be able to see how much money he's spent, where he's spent it and when. It will be displayed as a type of a bar chart, where the longer bar naturally represents the higher amount spent.
- **View Budget:**
This feature allows us to see how much our initial budget was, how much we've spent, and how much we have left.

Technology and Tools Involved

We turned to Java as the programming language because of its logic handling and modularity, which allows us to introduce subclasses rather than editing or deleting a chunk of code manually. Java's multi-threading capabilities allowed the app to store User data in the back-end, even when the app was closed by the user. Moreover, it's high-level nature allows the code to be readable. For the GUI, we used JavaFX to give a user-friendly experience to the client with vivid colors and buttons. We used file handling in the prototype to save data, but plan to use SQL databases in the future.

OOP Concepts Used

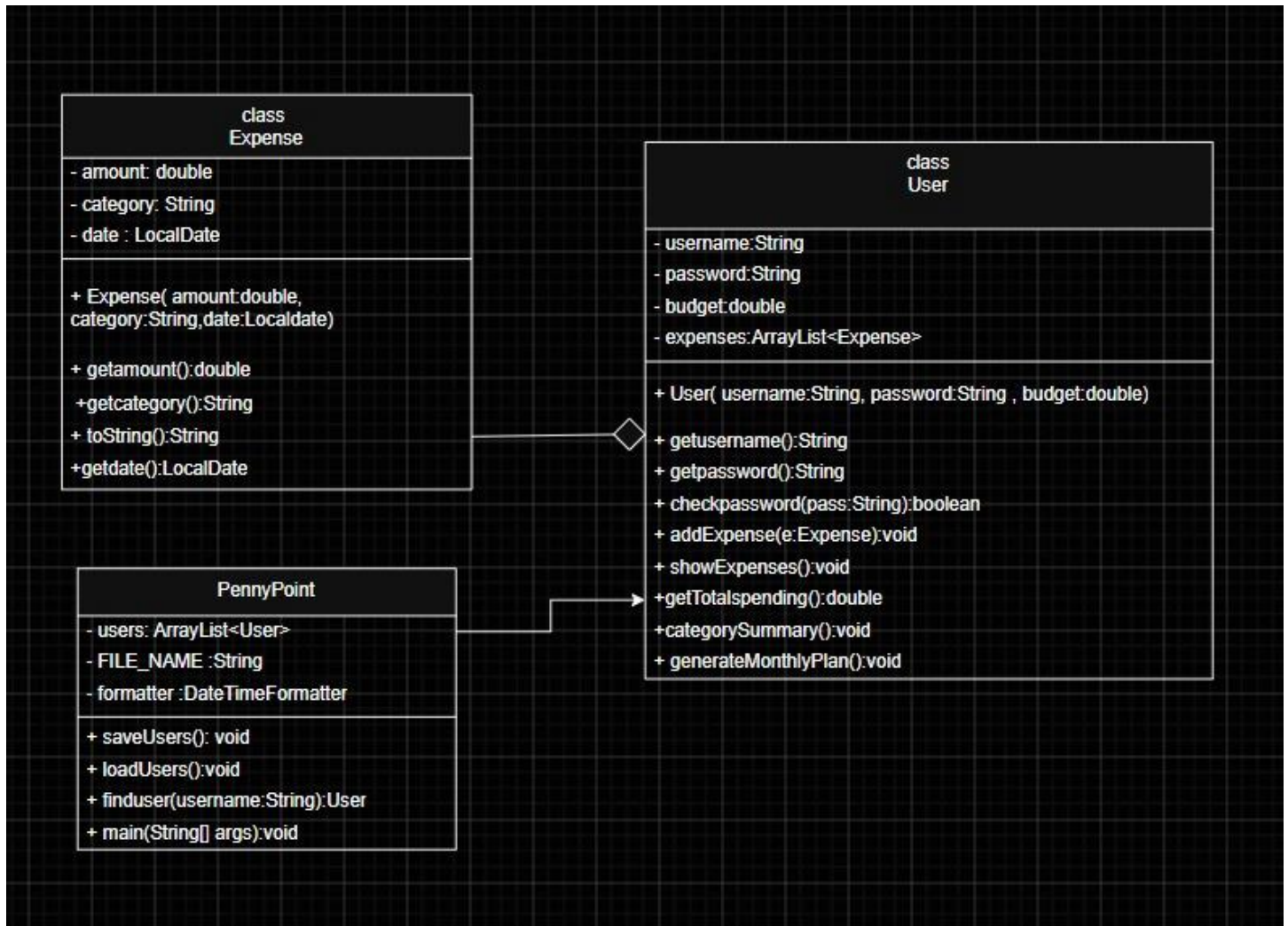
This app uses several OOP concepts like classes (User, Expense, PennyPoint), Inheritance (e.g PennyPoint extends Application), Interfaces (e.g User implements Serializable), Abstraction (Application being an abstract method), Method Overriding (e.g toString() in Expense class overrides Object's toString()) and Method Overloading (refreshExpenseList has different parameters).

We also see Serialization in User and Expense classes allowing data persistence and Composition when we see an ArrayList for Expense objects in the User class. Neither can exist without the other.

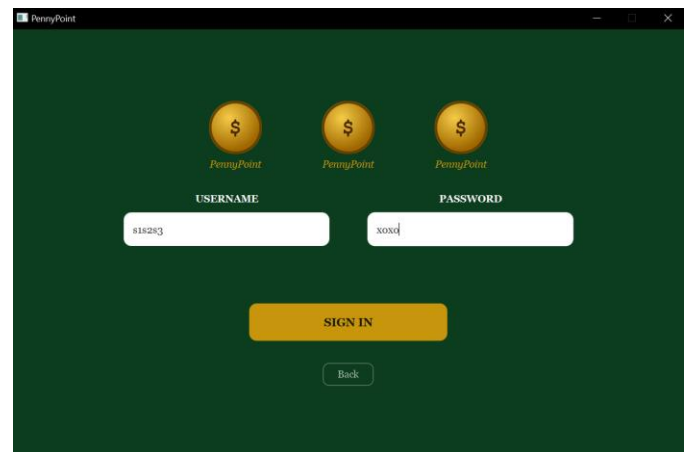
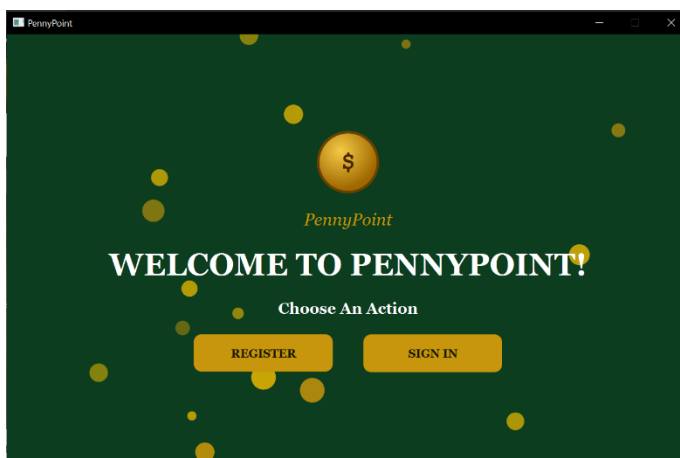
Team Members

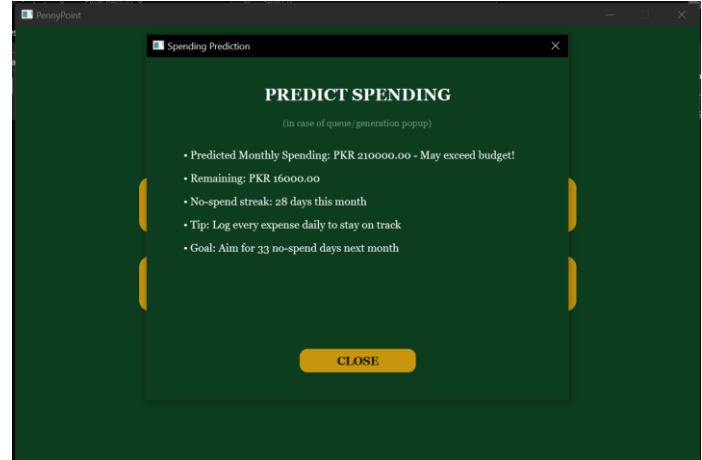
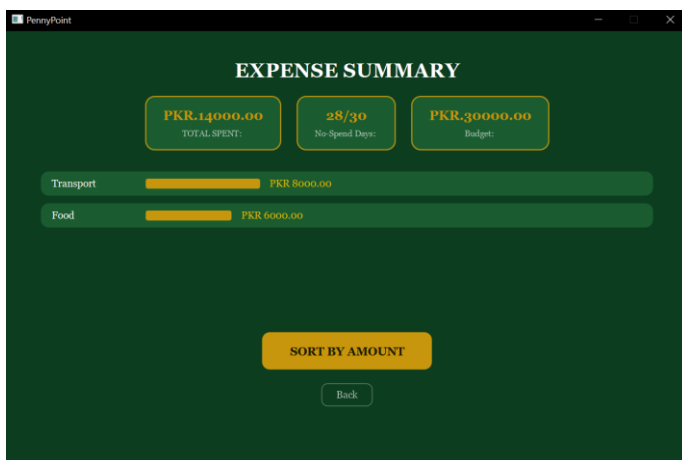
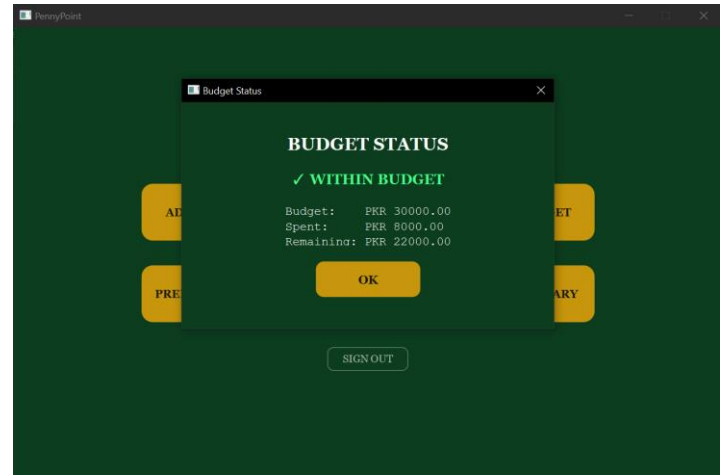
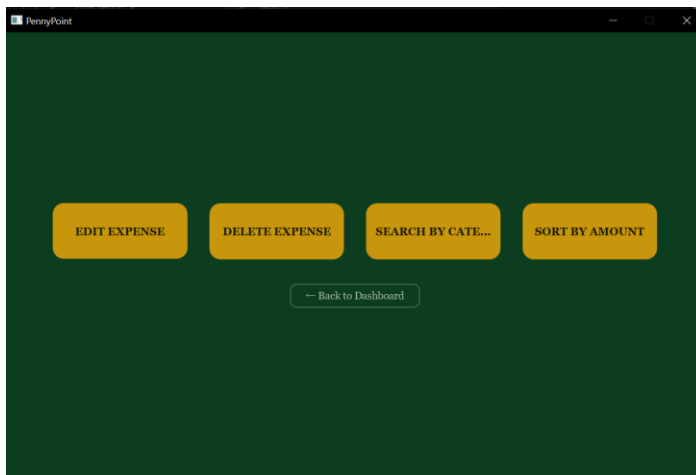
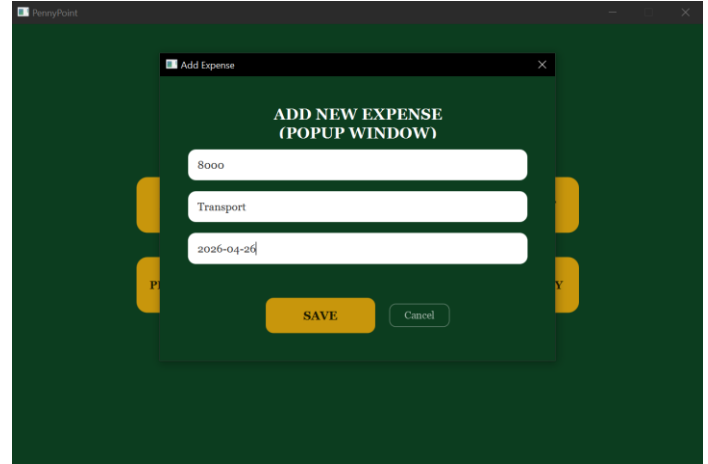
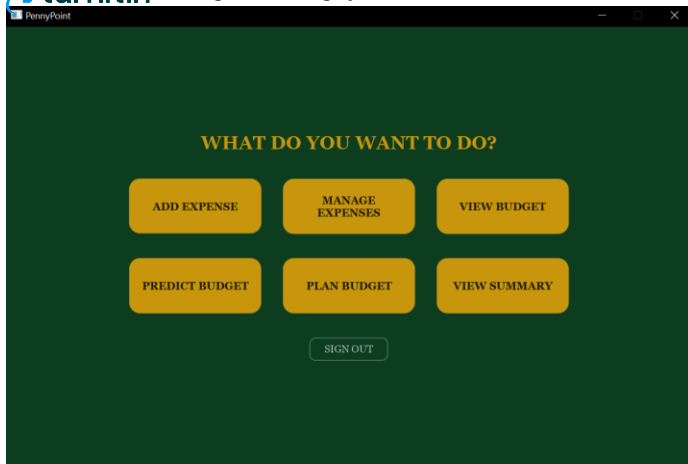
- **Muhammad Shahmeer Shaheedi:** He dealt with the Front-end of the app (GUI). He used Eclipse as the IDE and tried to make the app as easy to track and as interactive as possible given the time limit. He tried to make it centered around buttons for the sake of ease of use.
- **Maryam Ateeq:** She was responsible for the app's logic and Back-end. She initially created a whole console code with a User class, an Expense class, and a Main class, but in the GUI file, only the User and the Expense classes were needed.

UML DIAGRAM



SCREENSHOTS:





GITHUB LINK

<https://github.com/shahmeer0806/PennyPoint---565007---549392>